

Amazing Advances in 3D Printing And the Cause for Concern



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With a simple Google search, you can read about dozens of revolutionary developments in three-dimensional (3D) printing. You can read about how it has given wings to do-it-yourself (DIY) enthusiasts, how everybody is a maker today, how nutritious food can be 3D-printed in the shapes of cartoon characters to make it more attractive to kids, how an injured Winnipeg pooch called Oreo was made to run again using a 3D-printed knee cap, how a motorcycle accident survivor got a new 3D-printed face, how meat can be 3D-printed, and much more.

Let us look at five developments, which are certainly not as sensational as the ones you will find on your first Internet search, but are definitely very significant for the community.

Medicines, ready in a jiffy

Remember the days of yore, when doctors used to have various compounds ready in their clinic, and when patients came to them, they were given a mixture of relevant medicines, dutifully mixed by a compounder. For reasons of hygiene, absence of standards and the conveni-

ence of carry-along pills, the practice died down, especially in urban areas. Now, 3D printing seems ready to revive that concept, albeit in a more sophisticated way. A team of scientists headed by Dr Mohamed Albed Alhnan, School of Pharmacy and Biomedical Sciences, University of Central Lancashire, Preston, UK, has unveiled a way of 3D-printing low-cost, personalised medicines.

The basis for this innovation is a newly-developed drug-polymer filament system that can be used as the ink of a 3D printer. This can be used by doctors to print custom medicines in appropriate tablet sizes and weights. Doctors can control the dosage and form factors using simple software.

The team plans to bring the technology to hospitals and pharmaceutical firms within the next five years and to general public within the decade. They believe that this can greatly reduce the cost of manufacturing customised medicines for individuals and also help doctors treat ailments in a more personalised fashion. It will enable doctors to treat tough conditions using a trial-and-error approach and taper medication more systematically. As

time goes by, installing networked medicine printers at the homes of chronic patients can help doctors dole out customised individual doses.

Of course, a lot of standards and regulations will have to be in place before such medicine printers become available in the market, to safeguard patients and to instil best practices.

Dwellings for the homeless

In last year's technology focus on 3D printing (June 2014 issue), we looked at Kameron, a 3D-printing pavilion that can print segments of a room at one shot, making it easy to

Three-armed delta machine in action for 3D printing
(Courtesy: WASP)

